

Markowitz Approach on an portfolio optimization

Author: Macca Francesco

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1. Modern Portfolio Theory

1.1 Markowitz Mean-Variance Optimization

The purpose of this project is to implement and critically validate Markowitz's Modern Portfolio Theory, analyzing the difference between in-sample (optimization) and out-of-sample (real-world application) performance, to understand when and why the method works in practice.

Harry Markowitz's 1952 seminal work "Portfolio Selection" revolutionized investment management by introducing a mathematical framework for portfolio construction. His key insight was that investors should evaluate portfolios based on two dimensions: **expected return** (μ) and **risk measured by variance** (σ^2).

Core Principles:

Expected Portfolio Return:

$$E(R_p) = \sum w_i \times E(R_i)$$

1. where **w_i** represents the weight of asset i in the portfolio.
2. where **$E(R_i)$** represents the Expected Return of the asset

Portfolio Variance:

$$\sigma_p^2 = \sum \sum w_i \times w_j \times \sigma_{ij}$$

3. where σ_{ij} is the covariance between assets i and j .
4. **The Efficient Frontier:** The set of portfolios that maximize expected return for a given level of risk, or equivalently, minimize risk for a given level of expected return.

Sharpe Ratio Maximization:

$$\text{Sharpe} = (E(R_p) - R_f) / \sigma_p$$

5. Identifies the portfolio with the highest risk-adjusted return.

1.2 Diversification Benefit

Markowitz demonstrated that combining assets with less than perfect correlation reduces portfolio risk below the weighted average of individual asset risks. This mathematical property forms the foundation of diversification theory and remains central to modern investment practice.

2. Code Architecture

2.1 Implementation Overview

The analysis uses Python libraries such as NumPy, Pandas, SciPy to implement a comprehensive portfolio optimization framework. The code architecture consists of five main parts:

1: Data Acquisition and Preparation

- Downloads historical price data for 9 ETFs representing major asset classes
- Acquires dynamic risk-free rate (13-Week US Treasury Bill)
- Implements data quality checks and missing value treatment
- Period coverage: January 2005 to January 2025

2: Statistical Analysis

- Computes annualized returns, volatilities, and Sharpe ratios
- Calculates risk metrics, maximum drawdown, Value at Risk (95%)
- Assesses distribution properties: skewness and kurtosis
- Generates covariance matrix for optimization

3: Portfolio Optimization

- Implements three optimization strategies:
 - Max Sharpe: Maximizes risk-adjusted returns
 - Min Volatility: Minimizes portfolio variance
 - Risk Parity: Equalizes risk contributions across assets
- Applies realistic constraints: 0-40% per asset
- Uses Sequential Least Squares Programming (SLSQP) algorithm

4: Backtesting Engine

- Implements train-test split methodology:
 - Training: 2005-2020 (in-sample optimization)
 - Testing: 2021-2025 (out-of-sample validation)
- Simulates quarterly rebalancing
- Accounts for weight drift between rebalancing dates
- Calculates comprehensive performance metrics

5: Visualization and Analysis

- Generates efficient frontier with Monte Carlo simulation
- Produces six complementary analytical charts
- Enables comparative performance assessment

2.2 Asset Composition

The portfolio includes nine exchange traded funds (ETFs) covering major asset classes:

Ticker	Asset Class	Description
SPY	US Large Cap (top 500) Equity	S&P 500 Index
VGK	European Equity	FTSE Developed Europe
FEZ	Eurozone Equity	EURO STOXX 50
VWO	Emerging Markets	FTSE Emerging Markets
AGG	US Aggregate Bonds	Investment Grade Bonds
IEF	US Treasury Bonds	7-10 Year Maturity
GLD	Commodities	Gold
VNQ	US Real Estate	Vanguard REIT Index
IYR	US Real Estate	iShares REIT Index

(All assets demonstrated good data quality with less than 1% missing observations, ensuring robust statistical estimation.)

2.3 Risk-Free Rate Dynamics

This analysis uses the 13-Week US Treasury Bill rate, which varied significantly over the 20-year period:

- **Mean:** 1.58%
- **Standard Deviation:** 1.87%

- **Range:** Near-zero (2009-2021) to 5%+ (2006-2007, 2023-2024)

This dynamic approach provides more realistic Sharpe ratio calculations that reflect changing opportunity costs across different interest rate regimes.

3. Results and Analysis

3.1 Training Period Performance (2005-2020)

Asset-Level Statistics:

The training period revealed substantial performance dispersion across asset classes:

Asse	Annual Return	Volatility	Sharpe Ratio	Max Drawdown
AGG	4.26%	4.95%	0.61	-12.84%
IEF	5.04%	6.47%	0.59	-10.40%
GLD	10.57%	18.25%	0.51	-45.56%
SPY	11.02%	19.62%	0.50	-55.19%
VNQ	11.95%	31.20%	0.34	-73.07%

Observations:

1. **Bond Dominance in Risk-Adjusted Terms:** AGG and IEF achieved the highest Sharpe ratios (0.61 and 0.59) despite modest absolute returns. This reflects the 2005-2020 period's unique characteristics: two major equity crises (2008, 2020) and a secular decline in interest rates that benefited fixed income.
2. **Equity Returns with High Risk:** SPY delivered 11.02% annual returns but experienced a 55.19% maximum drawdown during the 2008-2009 financial crisis, highlighting the trade-off between long-term growth and short-term volatility.
3. **Distribution Anomalies:** The high kurtosis values (AGG: 83.04, SPY: 16.26) indicate fat-tailed return distributions, violating Markowitz's normality assumption and suggesting crisis risk is underestimated by standard deviation alone.

Optimized Portfolio Performance:

Strategy	Return	Volatility	Sharpe	Dominant Holdings
Max Sharpe	6.43%	5.03%	1.04	IEF 40%, AGG 31.9%, SPY 18.6%

Min Volatility	5.90%	4.63%	1.01	IEF 40%, AGG 40%
Risk Parity	6.40%	6.30%	0.82	AGG 36.2%, IEF 31.4%
Equal Weight	8.82%	16.24%	0.47	All assets 11.1%
SPY Only	11.02%	19.62%	0.50	SPY 100%

Insights:

1. **Optimization Success:** The Max Sharpe portfolio achieved a Sharpe ratio of 1.04, double that of SPY-only (0.50) and more than twice Equal Weight (0.47), demonstrating clear optimization benefit in the training period.
2. **Heavy Bond Allocation:** Optimal portfolios allocated 60-80% to bonds (AGG + IEF), reflecting bonds' superior risk-adjusted performance during a period dominated by declining rates and equity volatility.
3. **Diversification Penalty:** Equal Weight achieved 8.82% return versus Max Sharpe's 6.43%. This apparent contradiction reflects the "diversification penalty" when low-returning assets dilute high-performing ones—a key tension in portfolio construction.
4. **Risk Reduction Effectiveness:** Optimized portfolios reduced volatility to 4.63-6.30% versus Equal Weight's 16.24%, a reduction of 60-70%, validating Markowitz's diversification mathematics.

3.2 Test Period Performance (2021-2025): The Reality Check

The out-of-sample results reveal a dramatic reversal:

Strategy	Total Return	Annual Return	Volatility	Sharpe	Max DD
Max Sharpe	4.08%	1.01%	7.16%	-0.28	-17.79%
Min Volatility	-0.18%	-0.04%	6.96%	-0.44	-17.88%
Risk Parity	1.38%	0.34%	7.44%	-0.36	-19.41%
Equal Weight	19.31%	4.53%	11.68%	0.13	-24.56%
SPY Only	68.27%	13.95%	16.48%	0.66	-24.50%

Performance Degradation Analysis:

1. **Catastrophic Sharpe Collapse:**
 - Max Sharpe: 1.04 → -0.28 (127% decline)

- Min Volatility: 1.01 → -0.44 (144% decline)
- All optimized portfolios recorded negative Sharpe ratios

2. **SPY Outperformance:**

- SPY delivered 13.95% annual return, 13.8× the Max Sharpe portfolio
- SPY's 68.27% cumulative return dwarfed Max Sharpe's 4.08%
- Simple buy-and-hold beat sophisticated optimization decisively

3. **Equal Weight Resilience:**

- Equal Weight (19.31% total return) outperformed all optimized strategies
- Naive diversification proved superior to historical optimization

Root Cause Analysis:

A. Regime Shift: The Interest Rate Reversal

This 2021-2025 period witnessed a fundamental macro regime change:

- **2005-2020:** Declining rates (5% → 0%), benefiting bonds
- **2021-2025:** Rising rates (0% → 5%), sfavoriting bonds

The optimized portfolios' 60-80% bond allocations, optimal for the falling-rate training period, became a severe liability when rates reversed. AGG and IEF, which delivered positive returns in training, likely posted negative returns during 2022's bond bear market.

B. Equity Market Strength

Despite volatility (COVID crash, 2022 invasion of Ukraine), the 2021-2025 period saw robust equity returns driven by:

- Technology sector outperformance
- Post-COVID economic recovery
- AI investment boom

SPY's allocation in optimized portfolios (14.8-18.6%) was insufficient to capture this equity rally.

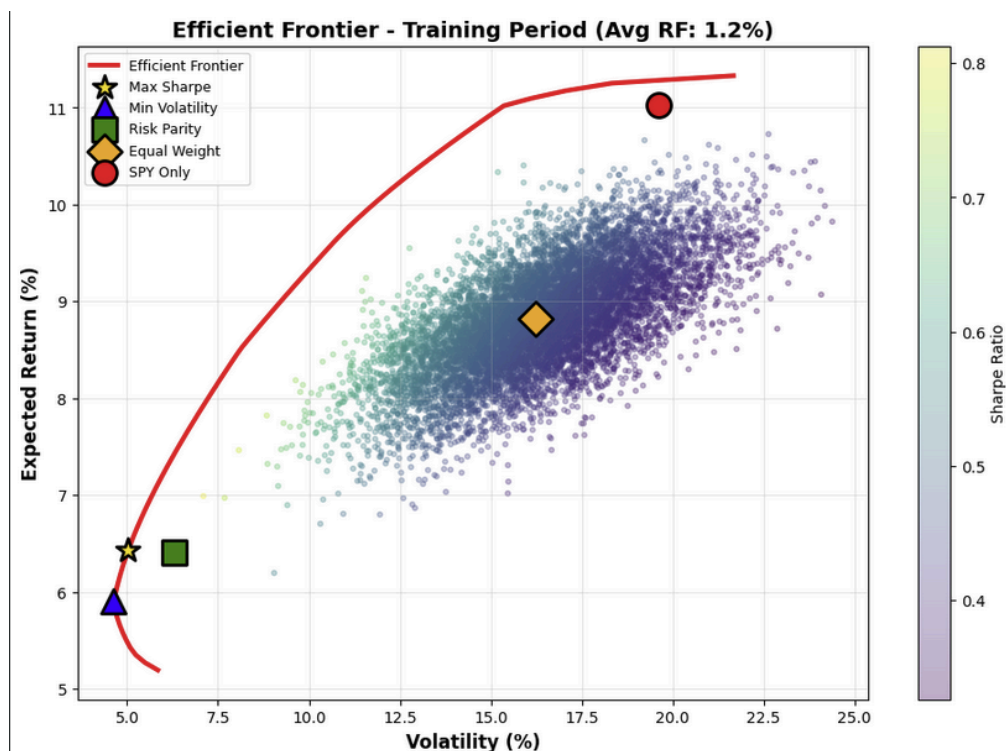
C. Estimation Error Amplification

This method (Markowitz optimization) is hypersensitive to input estimation errors. *Small errors in expected return estimates translate into large weight deviations.* The optimizer "overfit" to the training period's specific characteristics, failing to generalize to new market conditions.

3.3 Efficient Frontier Analysis

The efficient frontier visualization reveals the full spectrum of risk-return combinations:

1. **Training Period Frontier:** Shows classic convex shape with optimized portfolios positioned on or near the frontier, validating optimization mechanics.
2. **Monte Carlo Cloud:** The 10,000 random portfolios demonstrate that most non-optimized combinations are dominated, lying below the frontier.
3. **Strategic Positioning:**
 - Max Sharpe: Tangency point where the capital allocation line touches the frontier
 - Min Volatility: Leftmost point (global minimum variance portfolio)
 - Risk Parity: Interior position reflecting risk balancing rather than return maximization



3.4 Correlation Insights

The correlation matrix reveals key relationships:

1. **Equity Correlations:** SPY, VGK, FEZ, VWO shows correlations of "0.65-0.85", indicating globalization of equity markets limits diversification benefits within this asset class.
2. **Bond-Equity Correlation:** AGG and IEF show near-zero or slightly negative correlations with equities (typically -0.1 to +0.1), confirming bonds' traditional role as

equity hedges during risk-off periods.

3. **Independence of gold:** GLD exhibits low correlations (0.1-0.3) with most assets, supporting its portfolio diversification role.
 4. **REIT (Real Estate) Behavior:** VNQ and IYR correlate moderately with equities (0.5-0.6) and bonds (0.2-0.3), occupying a middle ground consistent with real estate's hybrid characteristics.
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4. Critical Limitations

4.1 Theoretical Limitations of Markowitz Framework

A. Normality Assumption

Mean-variance optimization assumes returns follow a normal distribution. The data contradicts this:

- Kurtosis values (AGG: 83.04, SPY: 16.26) indicate extreme fat tails
- Negative skewness in several assets suggests asymmetric downside risk
- **Implication:** Standard deviation understates true risk, particularly tail risk during crises

B. Single-Period Model

Markowitz optimization is fundamentally single-period: it optimizes for one investment horizon without considering:

- Dynamic rebalancing opportunities
- Time-varying investment opportunities
- Path-dependent factors (e.g., sequence-of-returns risk)

C. Historical Data Dependency

The model assumes future returns and covariances resemble historical patterns. This fails when:

- Structural regime shifts occur (interest rate reversals, regulatory changes)
- New market dynamics emerge (algorithmic trading, ETF proliferation)
- Black swan events disrupt historical relationships

D. Estimation Error

Expected return estimation is notoriously imprecise. Research shows:

- Mean return estimates have enormous standard errors over typical sample periods

- Covariance estimates are more stable but still subject to significant uncertainty
- Optimization magnifies these errors, producing unstable, concentrated portfolios

4.2 Data Limitations

A. The “*Survivorship Bias*”

All analyzed ETFs survived the entire 2005-2025 period. Funds that closed or merged are excluded, potentially overstating historical performance.

B. Short History

Twenty years of data includes only two major bear markets (2008, 2020). Longer histories could reveal different relationships.

C. ETF costs and issues

ETFs have tracking errors, expense ratios (0.03-0.15% annually), and tax considerations that differ from direct security ownership.

5. Markowitz in practice. (riguarda con Sara)

5.1 When Does Markowitz Work?

The analysis tells us Markowitz optimization adds value when:

1. **Stable Regimes:** Market relationships remain relatively constant
 2. **Long Horizons:** Short-term noise averages out over multi-decade periods
 3. **Risk Focus:** Investor prioritizes volatility reduction over absolute returns
 4. **Combined with Judgment:** Quantitative optimization is adjusted by qualitative market views
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6. Conclusions

The key conclusions are:

6.1 Theoretical Validation

The in-sample results validate Markowitz's core insights:

- Diversification mathematically reduces risk (60-70% volatility reduction achieved)
- Optimization can significantly improve risk-adjusted returns (Sharpe ratio doubled from 0.50 to 1.04)
- The efficient frontier exists as a practical tool for visualizing risk-return trade-offs

These findings confirm why Markowitz won the Nobel Prize: his framework provides a rigorous, mathematical foundation for portfolio construction.

6.2 Practical Challenges

The out-of-sample results reveal severe real-world limitations:

- Optimized strategies catastrophically underperformed in the test period (Max Sharpe: 1.01% vs SPY: 13.95%)
- Historical optimization failed to adapt to regime shifts (interest rate reversal)
- Simple strategies (SPY-only, Equal Weight) outperformed sophisticated optimization
- Negative Sharpe ratios indicate optimization destroyed value out-of-sample

These results underscore that theoretical soundness does not guarantee practical success.

6.3 The Core Tension

The analysis highlights the fundamental tension in quantitative portfolio management:

Optimization vs. Robustness

- Aggressive optimization (Max Sharpe) exploits historical patterns maximally but fails to generalize
- Conservative approaches (Equal Weight) sacrifice in-sample efficiency but prove more robust
- This bias-variance trade-off is central to all statistical learning problems

6.4 Forward-Looking Perspective

While i was testing this project I learned some important and critical lessons:

1. **Quantitative Skills Are Necessary But Insufficient:** Technical proficiency in optimization must be paired with economic intuition about markets, regimes, and structural change.
2. **Out-of-Sample Testing Is Mandatory:** In-sample results are misleading. Any investment strategy must be validated on data not used in its construction.
3. **Simplicity is not bad:** Complex models can underperform simple heuristics (for example a simple ETF on SP500).
4. **Risk Management Trumps Return Optimization:** The catastrophic out-of-sample failure stemmed from overconcentration in assets optimal for the training period. *Diversification across strategies, regimes, and approaches* matters more than optimizing within a single framework.
5. **The “Ceteribus Paribus” problem :** Markowitz provides the conceptual framework, but practical implementation requires awareness of assumptions, limitations, and

real-world frictions.

6.5 Final Assessment

Does Markowitz portfolio optimization "work"? The answer is "yes, but":

- **Yes, theoretically:** The mathematics are sound and the framework is invaluable for structured thinking about risk and return
- **Conditionally, practically:** Success depends on stable regimes, robust estimation techniques, and realistic constraints
- **No, naively:** Simply optimizing on historical data and implementing mechanically leads to poor outcomes

The value of this study lies not in proving Markowitz right or wrong, but in demonstrating the critical thinking required to bridge theory and practice. Understanding when and why models fail is as important as understanding when they succeed.

References

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Appendix

Software Environment:

- Python 3.8+
- NumPy 1.21+, Pandas 1.3+, SciPy 1.7+
- Matplotlib 3.4+, yfinance 0.1.70+
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Rebalancing the Simulation:

- Frequency: Quarterly (every 63 trading days)
- Reset to optimal weights
- Drift modeling: Daily weight updates between rebalancing

Statistical Conventions:

- Annualization factor: 252 trading days
 - Risk-free rate: Daily ^IRX converted to annual equivalent
 - Sharpe ratio: $(\text{Annual Return} - \text{Risk Free}) / \text{Annual Volatility}$
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